

Mainstreaming the Wrong Channel

Gender Policy, Adaptive Preferences, and the Limits of Redistribution

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Abstract

Three decades of gender mainstreaming have produced a paradox: unprecedented institutional commitment alongside persistent structural inequality. This paper argues that mainstream gender policy fails because it targets the wrong channel. The dominant approach—disaggregate, transfer, add a gender component—operates on the budget constraint: it redistributes resources to women. But the binding constraint on gender equality is informational, not allocative. Women receive systematically discounted signals about the returns to investment—through wage discrimination, occupational segregation, biased networks, and normative penalties for ambition. These signals reshape preferences across generations, producing lower aspirations, lower savings, and lower participation that appear voluntary but are endogenous to the signal structure. Drawing on three decades of practitioner critique, feminist economics, the behavioral aspirations literature, and recent formal work on adaptive preferences in general equilibrium, I develop a unified framework showing that: (i) lump-sum transfers cannot close preference-driven gaps because they do not alter the return signal; (ii) incremental reforms may have zero long-run effect if they fail to push the system past a tipping point; (iii) removing structural barriers produces hysteresis—preference gaps persist for multiple generations through cultural inertia; and (iv) standard welfare evaluation using women’s revealed preferences systematically understates the cost of inequality. The paper proposes five operational principles for signal-based mainstreaming: map the signal channel sector by sector, target the signal factor rather than the resource allocation, design for tipping points rather than incremental improvement, evaluate welfare against counterfactual preferences, and build feedback loops into program logic.

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1 Introduction

Gender mainstreaming was adopted as a global strategy at the 1995 Beijing Platform for Action and has since been institutionalized across virtually every multilateral and bilateral development agency. The World Bank, the United Nations system, the European Union, and most OECD governments have formal mainstreaming mandates requiring that gender perspectives be integrated into all stages of policy design, implementation, and evaluation. Thirty years later, the record is mixed at best. Sex-disaggregated data are now routinely collected. Gender focal points exist in most ministries. Project documents contain gender analyses. Yet the structural determinants of gender inequality—the wage gap, occupational segregation, the unequal distribution of unpaid care work, women’s underrepresentation in high-return sectors—have proven remarkably resistant to these interventions (Kabeer, 2005; Moser, 2005; Mukhopadhyay, 2004).

This paper argues that the persistence of gender inequality despite decades of mainstreaming effort is not primarily an implementation failure. It is a design failure rooted in a misdiagnosis of the binding constraint. The dominant mainstreaming paradigm treats gender gaps as a *resource allocation problem*: women lack money, skills, access, or representation, and the solution is to allocate more of these to them. The implicit theory of change is: transfer resources to women $\rightarrow$ women invest and participate more $\rightarrow$ gender gaps close. This logic operates entirely on what economists call the *budget constraint*—it changes what women can afford to do.

But gender inequality also operates through a second channel that mainstream policy largely ignores: the *signal channel*. Women receive systematically discounted signals about the returns to economic investment—through wage discrimination that reduces the observed return to education, through glass ceilings that truncate the observed return to career investment, through occupational segregation that concentrates women in low-return sectors, through biased information networks that exclude women from knowledge about high-return opportunities, and through social norms that penalize female ambition by imposing costs on women who deviate from expected roles (Goldin, 2014; Bertrand, 2011; Bursztyn et al., 2017). These discounted signals reshape women’s preferences across generations. When a girl observes that women in her community earn less than men for equivalent effort, her aspirations—her patience, her willingness to invest in her own future—adjust downward. Not because she is irrational, but because she is rationally incorporating a biased signal (Beaman et al., 2012; Bernard et al., 2014; Jensen, 2012).

The result is a feedback loop: structural barriers produce discounted return signals, which produce adapted preferences, which produce lower investment and participation, which re-

inforces the structural barriers. The loop is self-sustaining and, under certain conditions, self-amplifying. Standard policy—which redistributes resources but does not alter the signal structure—cannot break it. This is not a conjecture: recent formal work on adaptive preferences in general equilibrium demonstrates that lump-sum transfers are ineffective when the market failure operates through the informational channel rather than the budget constraint (Hernani-Limarino, 2025a), and that incremental signal improvements may have zero long-run effect if they fail to push the system past a tipping point in which rising female participation endogenously erodes discrimination (Hernani-Limarino, 2025b).

This paper synthesizes four bodies of literature—the practitioner critique of mainstreaming, feminist economics, the behavioral aspirations literature, and formal adaptive preference theory—into a unified framework for understanding why current gender policy fails and what an alternative would look like. The synthesis yields five operational principles for what I call *signal-based mainstreaming*: a reorientation of gender policy away from the resource channel and toward the informational mechanisms that drive preference formation.

The individual elements of this argument are not new. The practitioner critique of mainstreaming is two decades old (Razavi and Miller, 1995; Moser, 2005). The philosophical case that adaptive preferences undermine welfare evaluation dates to Sen (1985) and Nussbaum (2001), and has been applied specifically to women’s empowerment by Khader (2011), who argued that development practice systematically mistakes adapted preferences for genuine ones. The behavioral evidence that aspirations respond to signals is established (Ray, 2006; Lybbert and Wydick, 2018). And Jayachandran (2015) has provided a comprehensive synthesis of how social norms interact with economic returns to sustain gender inequality in developing countries.

What has been missing is a formal mechanism that connects these literatures—one that identifies the *precise channel* through which mainstreaming fails, proves that standard instruments cannot address it, and translates the diagnosis into operational policy design. This paper provides that mechanism, grounded in recent general equilibrium models of adaptive preference formation (Hernani-Limarino, 2025a,b). The contribution relative to existing work is threefold. First, relative to the philosophical critique (Sen 1985; Nussbaum 2001; Khader 2011): the paper embeds adaptive preferences in a production economy with endogenous returns, proving formally that the Second Welfare Theorem fails—that no schedule of lump-sum transfers can implement the first-best allocation when preferences co-evolve with prices. This moves the critique from “revealed preferences are problematic” to “here is the exact parameter (λ) that fiscal policy cannot reach, and here is the proof.” Second, relative to Jayachandran (2015): her framework identifies social norms and economic incentives as interacting determinants of gender inequality, but does not formalize the feedback from

economic outcomes to preference formation. The adaptive preferences framework closes the loop: norms shape signals, signals shape preferences, preferences shape investment, investment shapes returns, and returns shape the next generation’s norms. It is this feedback structure that generates the tipping points, hysteresis, and welfare mismeasurement that are absent from her analysis. Third, relative to the aspirations literature ([Genicot and Ray 2017](#); [Ray 2006](#)): the paper translates aspirational dynamics from a poverty-trap framework into a gender mainstreaming framework, showing that the same logic that generates aspiration traps at the individual level generates signal-driven preference traps at the group level, with specific implications for policy design (concentration vs. diffusion, generational time horizons, counterfactual welfare evaluation).

Section 2 documents the mainstreaming problem through three diagnostic lenses. Section 3 presents the adaptive preferences framework that unifies the diagnoses, including the distinction between informational and structural sources of signal discount. Section 4 identifies three specific mechanisms through which current policy fails. Section 5 develops five operational principles for signal-based mainstreaming. Section 6 provides a systematic comparison of the two paradigms. Section 7 develops detailed applications across six sectors, opening with the care economy as a cross-cutting signal amplifier. Section 8 concludes.

2 Three Diagnoses of a Known Problem

The failure of gender mainstreaming to produce structural change has been recognized from within the development community for at least two decades. The diagnoses, however, differ in their depth and implications.

2.1 The Practitioner Diagnosis: Implementation Failure

The dominant institutional narrative attributes mainstreaming’s limitations to insufficient implementation. On this account, the strategy is correct but the execution is weak: too few resources, too little political will, too much bureaucratic dilution. [Moser \(2005\)](#) documented how gender mainstreaming across UN agencies was reduced to a compliance checklist—sex-disaggregated indicators, gender-sensitive language in project documents, the occasional training workshop—with no connection to the causal logic of the programs themselves. [Mukhopadhyay \(2004\)](#) showed how mainstreaming paradoxically made gender *less* visible by diffusing responsibility: when gender is “everyone’s job,” it becomes no one’s job. [Staudt \(2003\)](#) documented how USAID’s mainstreaming mandate became a bureaucratic ritual in which gender analyses were conducted to satisfy approval requirements and then filed away,

with no influence on project design or implementation. [Walby \(2005\)](#) analyzed the European experience and found a persistent gap between the transformative rhetoric of mainstreaming policy documents and the incremental, technocratic reality of their implementation.

The practitioner diagnosis is accurate but insufficient. It correctly identifies that mainstreaming has been poorly implemented, but it assumes that better implementation of the same strategy would produce different results. The implicit claim is that the theory of change—transfer resources and access to women, and gender gaps will close—is correct; the problem is merely operational. The next two diagnoses challenge this assumption.

2.2 The Feminist Economics Diagnosis: Wrong Framework

A deeper critique, developed primarily within feminist economics, argues that the problem is not implementation but the underlying analytical framework. Standard economics treats preferences as exogenous, households as unitary, and markets as gender-neutral. Gender mainstreaming built on this framework inherits its blind spots.

[Sen \(1985, 1990\)](#) made the foundational argument that the preferences of the disadvantaged cannot be taken at face value for welfare evaluation. The “deprived tend to come to terms with their deprivation,” adjusting desires to what they “unambitiously see as feasible” ([Sen, 1985](#), p. 22). Applied to gender, this means that women’s observed choices—lower investment in education, earlier labor market exit, concentration in lower-paid occupations—may reflect adapted preferences rather than genuine welfare-maximizing decisions. [Nussbaum \(2001\)](#) developed this into the capabilities approach, arguing that welfare should be evaluated against what people are *able* to do and be, not against what they have been conditioned to want. [Elson \(1995\)](#) demonstrated how macroeconomic policy systematically ignores gendered impacts because the models it uses assume away intrahousehold inequality and unpaid care work. [Agarwal \(1997\)](#) showed that intrahousehold bargaining models reveal stark inequalities hidden by the unitary household assumption. [Folbre \(1994\)](#) argued that women’s economic “choices” are made under structural constraints so severe that the language of free choice is misleading.

This diagnosis is closer to correct. It identifies the core problem: mainstream economics—and the policy built on it—treats women’s observed behavior as reflecting stable, exogenous preferences, when in fact those preferences are shaped by the very inequality that policy aims to address. [Khader \(2011\)](#) developed this critique into a systematic argument about women’s empowerment in development practice, showing that adaptive preferences are routinely mistaken for “cultural values” or “informed choices” in program design and evaluation. Her work demonstrates that the philosophical point has direct operational consequences: if

a program takes women’s low educational aspirations as a baseline rather than as a symptom, it will set targets that accommodate the adaptation rather than challenge the signal structure that produced it.

Yet the feminist economics critique, including Khader’s careful treatment, has lacked a formal mechanism connecting signal distortion to preference formation to welfare loss within an equilibrium framework. [Jayachandran \(2015\)](#) provided a comprehensive synthesis of how social norms and economic incentives interact to sustain gender inequality, but her framework—like most in the literature—treats norms and preferences as slow-moving background variables rather than as endogenous objects whose dynamics can be formally characterized. Without a model that specifies *how* signals shape preferences, *how fast* adapted preferences converge or diverge, *whether* the system exhibits multiple equilibria, and *which* policy instruments can and cannot shift the equilibrium, the critique remains a challenge to the standard approach rather than an operational alternative. The framework presented in [Section 3](#) provides the missing mechanism.

2.3 The Behavioral Diagnosis: Preferences Adapt

The third strand of evidence, emerging from behavioral and development economics over the past fifteen years, provides the micro-level empirical foundation. A growing body of experimental and quasi-experimental work demonstrates that economic preferences—aspirations, patience, risk tolerance, investment behavior—respond causally to environmental signals, and that these responses are persistent across time and generations.

[Beaman et al. \(2012\)](#) exploited the random assignment of female leaders to Indian village councils to show that exposure to women in positions of authority durably raised girls’ aspirations and educational attainment. The effect operated not through resources (the councils controlled no education budget) but through the signal: seeing women lead changed what girls believed was possible. [Bernard et al. \(2014\)](#) conducted aspirations interventions in rural Ethiopia, showing documentaries about individuals who overcame poverty. Five years later, treated individuals had higher aspirations, greater investment, and better economic outcomes—from a one-hour signal intervention with no resource transfer. [Jensen \(2012\)](#) provided Dominican adolescents with accurate information about the returns to education and observed significant increases in schooling—a direct correction of the discounted signal ($\lambda \rightarrow 1$) with no accompanying transfer.

On the negative side, [Bursztyn et al. \(2017\)](#) demonstrated that single MBA women at an elite U.S. business school strategically downgraded their stated career ambitions when they believed male classmates would observe their responses—direct evidence of social norms

operating as a real-time signal discount. [Malmendier and Nagel \(2011\)](#) showed that individuals who experienced low stock market returns during their formative years remained more risk-averse decades later, even controlling for current wealth, demonstrating persistent preference adaptation in response to experienced return signals. [Giuliano and Spilimbergo \(2014\)](#) established that growing up during a recession shifts beliefs about the role of luck versus effort for an entire generation.

At the theoretical level, [Ray \(2006\)](#) laid the conceptual groundwork by arguing that aspirations—shaped by the economic outcomes of one’s reference group—mediate the relationship between inequality and investment, generating “aspiration failure” among the disadvantaged. [Genicot and Ray \(2017\)](#) formalized this, showing how socially determined aspirations can generate multiple steady states and aspiration traps. [Bowles \(1998\)](#) argued that markets and institutions shape preferences, not just the reverse. [Doepke and Zilibotti \(2017\)](#) showed that parenting styles—and hence the preferences transmitted to children—respond optimally to economic incentives. [Bernheim et al. \(2021\)](#) developed a theory of “chosen preferences” in which agents select worldviews consistent with their experience, providing a microfoundation for adaptive updating. The thread connecting these contributions is clear: preferences are endogenous, they respond to economic signals, and the response has dynamic consequences. What remains undone is the translation from “preferences adapt” to “here is why gender mainstreaming fails and here is what to do instead.”

Taken together, these three diagnoses point toward a common conclusion: gender mainstreaming fails because it treats preferences as fixed and operates exclusively on the budget constraint, while the binding constraint is informational—it resides in the signals that shape preferences. Each diagnosis captures part of the picture: the practitioners see the symptom (programs don’t work), the feminist economists identify the structural cause (the framework assumes away the problem), and the behavioral economists provide the micro evidence (preferences demonstrably adapt to signals). The next section provides a formal framework that unifies the three diagnoses into a single analytical structure with testable implications and operational prescriptions.

3 The Adaptive Preferences Framework

I draw on recent work embedding endogenous preferences in a production economy ([Hernani-Limarino, 2025a](#)) and its application to gendered inequality ([Hernani-Limarino, 2025b](#)) to present the core mechanism in accessible terms. The full formal treatment, including proofs, appears in those companion papers; here I state the model’s structure and its four key implications for policy. The framework’s value-added relative to existing treatments—[Sen’s](#)

philosophical critique, [Khader](#)’s empowerment analysis, [Genicot and Ray](#)’s aspiration-trap model, and [Jayachandran](#)’s norm-incentive synthesis—is that it embeds all of these insights in a single general equilibrium structure with heterogeneous return signals, generating results that are simultaneously formal (provable), dynamic (with explicit convergence rates and tipping points), and operational (identifying which policy instruments can and cannot shift the equilibrium).

3.1 The Mechanism

Consider an economy with two groups of agents—call them “high-access” (H) and “low-access” (L)—who differ in one respect: the return signal they observe. H -agents observe the true market return on investment, r_t . L -agents observe a discounted signal, λr_t , where $\lambda \in (0, 1]$ captures the aggregate effect of all mechanisms—discrimination, segmentation, informational barriers, normative penalties—that depress the returns women perceive or realize.

Agents have a “patience parameter” θ that governs their willingness to invest in the future (save, acquire education, build a business). This parameter is not fixed; it updates across generations in response to experienced returns:

$$\theta_{t+1}^g = \rho \theta_t^g + (1 - \rho)\bar{\theta} + \mu(\tilde{r}_t^g - r^*), \quad (1)$$

where $\rho \in [0, 1]$ captures cultural persistence (how much children’s aspirations resemble their parents’), $\bar{\theta}$ is a societal baseline, $\mu > 0$ is the speed of adaptation, and r^* is a reference return. When the signal $\tilde{r}_t^g$ exceeds the reference, agents become more patient (invest more); when it falls short, patience declines.

The economy closes through a production technology and competitive factor markets: aggregate investment determines the capital stock, which determines the return r_t , which feeds back into preference dynamics. The system is a feedback loop:

$$\lambda \longrightarrow \theta^L \longrightarrow \text{savings} \longrightarrow K \longrightarrow r \longrightarrow \lambda r \longrightarrow \theta^L \longrightarrow \dots$$

Under log utility, optimal saving takes the closed form $s = \theta/(1 + \theta)$, making the entire dynamical system analytically tractable.

3.2 Two Sources of Signal Discount

A conceptual distinction is important for policy. The signal discount $\lambda < 1$ can arise from two different sources, and the policy response differs accordingly.

The first source is *informational*: women receive less or worse information about available returns. A woman excluded from a male-dominated professional network does not know about a high-return investment opportunity that her male counterpart knows about. A girl whose school counselor steers her away from STEM fields never learns what the return to engineering is. In these cases, the signal is *biased*—the true return to female investment is r_t (the same as for men), but women observe a discounted version λr_t because of informational barriers. The appropriate intervention is information correction: provide accurate signals, and the discount disappears.

The second source is *structural*: women genuinely earn lower returns on equivalent investments due to discrimination, segmentation, or normative penalties. A woman with the same degree as a man earns less because of wage discrimination. A female farmer growing the same crop receives a lower price from a biased intermediary. In these cases, the signal is *accurate*—the return to female investment genuinely is λr_t , and women’s adapted preferences are a rational response to a real constraint. The appropriate intervention is structural reform: change the return itself, not just the information about it.

In practice, most settings involve both channels simultaneously. A woman may face a genuine wage gap (structural) *and* lack information about the sectors where the gap is smallest (informational). The compound discount is $\lambda = \lambda_{\text{info}} \times \lambda_{\text{struct}}$, and a complete signal-based intervention must address both components. Information correction alone—telling women about returns that are genuinely lower—may raise aspirations temporarily but will produce aspiration frustration when the structural reality fails to match the corrected signal. Structural reform alone—equalizing returns without correcting the informational lag—will be slow to affect preferences because women may not observe the change. The sectoral applications in Section 7 trace both channels in each domain.

3.3 Four Key Results

The formal analysis ([Hernani-Limarino, 2025a,b](#)) establishes four results with direct implications for gender policy.

Result 1: Endogenous Preference Divergence. Starting from identical initial conditions ($\theta_0^H = \theta_0^L$), the preference gap $D_t = \theta_t^H - \theta_t^L$ is strictly positive from period 1 onward and converges to a steady state:

$$D^* = \frac{\mu(1 - \lambda)r^*}{1 - \rho} > 0. \quad (2)$$

Women’s lower aspirations are not a cultural residual or a pre-existing condition; they are produced endogenously by the signal discount $\lambda < 1$ interacting with adaptive preference formation. The gap increases in the signal discount $(1 - \lambda)$, the adaptation speed (μ) , and cultural persistence (ρ) .

Result 2: Transfer Ineffectiveness (Second Welfare Theorem Failure). No schedule of lump-sum transfers from H to L agents can close the preference gap. The formal proof is simple: the preference update rule (1) depends on the return signal $\tilde{r}_t^g$, not on disposable income. A transfer changes what women can spend but not what they observe about the returns to investment. The gap recursion $D_{t+1} = \rho D_t + \mu(1 - \lambda)r_t$ does not contain the transfer τ_t . This is a failure of the Second Welfare Theorem: the standard tool of redistributive policy cannot implement the efficient allocation when the market failure is informational rather than allocative.

Result 3: Tipping Points and Multiple Equilibria. When the signal discount is itself endogenous—when discrimination weakens as women’s participation rises, so that $\lambda = \lambda(\theta^L)$ is an increasing function—the system can exhibit multiple steady states: a “low-participation trap” in which low aspirations sustain low participation which sustains high discrimination, and a “high-participation equilibrium” in which high aspirations sustain high participation which erodes discrimination. Between them lies a tipping point: a critical level of female aspirations above which the feedback becomes virtuous and below which it remains vicious. Incremental reforms that move the system within the basin of attraction of the low trap have zero long-run effect; only reforms that push past the tipping point trigger self-sustaining change (Hernani-Limarino, 2025b).

Result 4: Hysteresis. Even after the signal discount is fully eliminated ($\lambda \rightarrow 1$), the preference gap decays geometrically at rate ρ^n per generation. The half-life is $\ln 2 / |\ln \rho|$ generations. For cultural persistence of $\rho = 0.85$ —a plausible value given the intergenerational evidence on preference transmission—the half-life is approximately 4.3 generations, or roughly 100 years. Structural reform that “removes barriers” at time T does not produce equality at time $T + 1$; it initiates a multi-generational convergence process whose speed depends on how deeply preferences have adapted.

4 Three Mechanisms of Policy Failure

The framework identifies three specific mechanisms through which current mainstreaming fails. Each corresponds to a mismatch between the policy instrument and the nature of the constraint.

4.1 Treating Adapted Preferences as Welfare-Relevant

Cost-benefit analyses of gender programs typically use women’s observed behavior as the welfare benchmark. If women “choose” not to invest in higher education, to exit the labor force after marriage, or to concentrate in low-return occupations, the evaluation takes these choices as reflecting genuine preferences and measures program impact against them. A program that marginally improves women’s outcomes relative to this baseline is judged successful.

But if preferences are endogenous, this methodology is circular. Women’s “revealed preferences” reflect adapted θ^L —the patience parameter that has been depressed by a lifetime (and multiple generations) of discounted return signals. Evaluating welfare with adapted preferences is equivalent to measuring the cost of poverty using the aspirations of the poor: by construction, it understates the loss, because the metric has adjusted to accommodate the deprivation.

The correct welfare comparison is between outcomes under adapted preferences and outcomes under counterfactual baseline preferences $\bar{\theta}$ —what women would prefer absent the signal distortion. In the formal model, this gap is quantifiable: the difference between $U(c^y, c^o; \theta^L)$ and $U(c^y, c^o; \bar{\theta})$ measures the “welfare illusion”—the degree to which endogenous preference adaptation masks true deprivation. For plausible calibrations, this mismeasurement is substantial: women appear to lose less welfare than they actually do, precisely because the signal structure has taught them to want less ([Hernani-Limarino, 2025a](#)).

This connects directly to the philosophical critique of [Sen \(1985\)](#) and [Nussbaum \(2001\)](#), but gives it a quantitative foundation. The policy implication is immediate: any evaluation framework that takes women’s observed choices as the welfare benchmark will systematically understate the cost of gender inequality and overstate the adequacy of current interventions.

4.2 Ignoring Hysteresis

Development programs typically operate on 3–7 year time horizons. Gender interventions within these programs—quotas, training programs, affirmative action measures—are designed to raise women’s participation and outcomes within the project cycle. Many of these

interventions temporarily raise λ toward 1 during the program period: a quota ensures women are hired; a training program provides access to male-dominated skills; a credit line offers equal terms.

But the preference gap has a half-life measured in generations, not project cycles. A program that raises λ from 0.70 to 0.90 for five years and then ends will observe the preference gap partially close during implementation and then reopen afterward, because adapted preferences ($\theta^L < \theta^H$) persist through cultural transmission even after the structural intervention is withdrawn. Evaluators will conclude that “the intervention had no sustained impact.” The correct conclusion is: the intervention was too short to shift the preference equilibrium.

This mechanism explains a recurring puzzle in the gender and development literature: programs that produce significant short-run effects on women’s outcomes but show fade-out in longer-run follow-ups (Banerjee et al., 2015). The standard interpretation is that the program’s causal mechanism was weak or temporary. The adaptive preferences interpretation is different: the program’s effect on the signal channel (λ) was real but the induced preference change had not yet propagated far enough to become self-sustaining. The distinction has profound implications for program design: it argues for permanent institutional reform over temporary project-based interventions, and for evaluation horizons measured in decades rather than project cycles.

4.3 Missing the Tipping Point

The most consequential failure of current mainstreaming may be its diffusion of effort. Mainstreaming mandates typically require that *all* programs in *all* sectors incorporate gender considerations. In practice, this means spreading limited resources across many small interventions: a gender component in the road project, a women’s session in the agricultural extension program, a sex-disaggregated indicator in the health monitoring system. Each intervention produces a modest, incremental improvement in λ within its narrow domain.

If the system exhibits multiple equilibria—as the formal analysis with endogenous λ suggests—incremental improvements are effective only if they push the system past the tipping point that separates the low-participation trap from the high-participation equilibrium. Below the tipping point, feedback is negative: any improvement in women’s participation that is not large enough to significantly erode discrimination (λ) is reversed by the persistence of adapted preferences. Above the tipping point, feedback turns positive: rising participation erodes discrimination, which raises signals, which raises aspirations, in a self-sustaining virtuous cycle.

The “add women and stir” approach to mainstreaming—many small interventions spread across the institutional landscape—is precisely the strategy least likely to reach a tipping point. It is the policy equivalent of trying to bring water to a boil by turning up the heat 0.1 degrees in twenty different pots rather than bringing one pot to 100°C.

5 Five Principles for Signal-Based Mainstreaming

The adaptive preferences framework suggests a fundamental reorientation of gender mainstreaming: from the resource channel (transfers, quotas, access) to the signal channel (the informational mechanisms that shape preferences). I propose five operational principles.

5.1 Principle 1: Map the Signal Channel

Before designing any gender intervention, identify how $\lambda < 1$ operates in the specific sector. This means asking: through what mechanisms do women receive systematically discounted signals about the returns to investment?

The signal channel is sector-specific. In agriculture, λ may operate through gender-segmented output markets: women produce the same crops but sell through intermediaries who offer lower prices, or lack access to extension services that convey information about new technologies (World Bank, 2012). In finance, λ operates through discriminatory pricing: women face higher effective interest rates, shorter loan terms, or exclusion from high-return asset classes (Demirgüç-Kunt et al., 2013). In labor markets, λ operates through occupational segregation: women observe that female-dominated occupations pay less for equivalent human capital, depressing the perceived return to education and experience (Goldin, 2014). In education, λ operates through the downstream labor market: if employers discount women’s qualifications, the perceived return to schooling is lower for girls than boys (Jensen, 2012).

The diagnostic question is not “are there gender gaps?” (there always are) but “where does λ enter the causal chain, how large is the discount, and what institutional mechanisms produce it?” This requires sector-specific analysis that goes beyond sex-disaggregated outcome statistics to trace the *information structure* facing men and women in each domain.

5.2 Principle 2: Target λ , Not Just w

Once the signal channel is identified, interventions should raise λ directly rather than (or in addition to) transferring resources.

Information interventions provide women with accurate return signals, correcting the discount. Jensen (2012) demonstrated that providing accurate information about returns to

education shifted girls’ schooling decisions in the Dominican Republic—a direct λ -correction with no resource transfer. [Bernard et al. \(2014\)](#) showed that a low-cost aspirations intervention in Ethiopia produced effects lasting at least five years. These are proof-of-concept demonstrations that targeting the signal is feasible and effective.

Market access interventions connect women to the same markets men use. Digital platforms that enable direct-to-buyer agricultural sales eliminate intermediary discrimination, raising λ by removing the agent who imposed the discount. Mobile banking that provides equal-terms financial access raises the return signal for female savers.

Institutional reform directly raises λ through legal and regulatory channels: equal pay legislation, blind hiring mandates, anti-discrimination enforcement, transparency requirements on gender pay gaps. These are λ -raising policies that change the signal women observe about the returns to their investment.

Algorithmic auditing addresses a new frontier. [Lambrecht and Tucker \(2019\)](#) showed that ad-targeting algorithms systematically show high-paying career advertisements to men over women, even when the advertiser seeks gender balance. If the information delivery systems of the 21st century embed $\lambda < 1$, auditing and correcting them is a signal intervention with potentially large reach.

5.3 Principle 3: Design for Tipping Points

The multiple equilibria result argues against diffusion and for concentration. If there exists a tipping point below which the feedback is vicious and above which it is virtuous, policy should concentrate effort to cross the threshold rather than spread marginal improvements.

Saturation interventions apply this logic. Instead of training 10% of women farmers in each of 50 districts, train 80% in 5 districts. If $\lambda(\theta^L)$ is endogenous—if discrimination weakens as female participation visibly rises—then reaching critical mass in a few locations triggers the self-sustaining virtuous cycle. [Beaman et al. \(2012\)](#) provide evidence consistent with this: exposure to female leaders needed to reach a threshold of two consecutive terms before aspirational effects became durably detectable.

Sequencing exploits complementarities between signal correction and aspiration shocks. The formal analysis shows that aspiration interventions (role models, mentoring, media exposure—which shock θ^L upward) combined with structural reform (which raises λ) outperform either intervention alone ([Hernani-Limarino, 2025b](#)). The optimal sequence leads with aspiration shocks to build demand, then implements structural reform to validate and sustain the new aspirations. Neither works alone: aspiration shocks without structural change produce temporarily raised hopes that the signal structure crushes; structural change without

aspiration work produces opportunities that adapted preferences fail to exploit.

Duration must match hysteresis. If the preference half-life under cultural persistence of $\rho = 0.85$ is approximately four generations, a five-year program is structurally insufficient. This argues for permanent institutional change (constitutional provisions, legal reform, permanent regulatory bodies) over temporary project-based interventions. Where project-based interventions are unavoidable, they should be designed to create self-sustaining local institutions that maintain the signal correction after the project exits.

5.4 Principle 4: Evaluate Welfare Correctly

The welfare mismeasurement result demands a change in evaluation methodology. Current practice evaluates gender programs against women’s observed choices (adapted preferences). Signal-based mainstreaming requires evaluation against counterfactual preferences.

Three specific methodological shifts follow.

Counterfactual welfare benchmarks. Evaluate program impacts against what women’s welfare *would be* if they valued the future at the baseline rate $\bar{\theta}$ rather than the adapted rate θ^L . The gap between adapted and counterfactual welfare measures the true cost of the signal failure. In practice, this can be approximated by using men’s aspiration levels (in the same community and economic stratum) as the counterfactual for women’s, under the assumption that men’s aspirations are less distorted by the signal discount.

Process metrics. Track λ directly, not just outcomes. Measure the gender gap in perceived returns (through aspirations surveys), in actual returns (through administrative earnings data), and in signal exposure (through information access audits). If λ is not moving, the program is not mainstreaming—it is transferring.

Generational evaluation horizons. Accept that the relevant time horizon for gender mainstreaming is generational, not annual. Build intergenerational tracking into monitoring and evaluation frameworks: measure aspiration transmission (θ inheritance) across cohorts, not just contemporaneous outcomes. The [Bernard et al. \(2014\)](#) five-year follow-up is a model; what is needed is a 20-year follow-up that tracks the children of treated individuals.

5.5 Principle 5: Make the Feedback Loop Explicit

Current mainstreaming treats gender as a “cross-cutting theme” to be “integrated” into sectoral programs. The framework developed here says it should be treated as a dynamic system with endogenous feedback. The program’s theory of change must include the feedback loop, not just the first-round effect.

A standard logframe for a gender intervention reads:

Output: 500 women trained $\rightarrow$ Outcome: 20% increase in female income.

This is a one-shot, budget-constraint theory of change. It assumes preferences are fixed: give women skills, they invest them, income rises. It ignores the feedback from income to signals to preferences to behavior.

A signal-based theory of change reads:

Output: Signal correction (λ raised from 0.70 to 0.85 in target sector) $\rightarrow$ Intermediate outcome: Aspiration adjustment (θ^L increases by x within cohort) $\rightarrow$ Long-run outcome: Self-sustaining participation increase (λ endogenously exceeds tipping point).

The key differences: the output is a change in the signal structure, not a count of beneficiaries; the intermediate outcome is a change in preferences, not just income; and the long-run outcome is a phase transition in the system's dynamics, not a marginal improvement in a flow variable. This requires new indicators (aspiration levels, perceived returns, signal exposure), new time horizons (generational tracking), and a new concept of success (crossing the tipping point, not maximizing the number of beneficiaries).

6 Standard vs. Signal-Based Mainstreaming: A Systematic Comparison

Before turning to sectoral applications, it is useful to lay out the structural contrast between the two approaches. They differ not only in which interventions they recommend but in how they conceptualize the problem, what they measure, and what they consider success. Table 1 summarizes the comparison across eight dimensions.

Table 1: Two Paradigms of Gender Mainstreaming

Dimension	Standard (Resource-Based)	Signal-Based
Root diagnosis	Women lack resources, access, representation	Women receive discounted return signals that reshape preferences
Theory of change	Transfer resources $\rightarrow$ women invest $\rightarrow$ gaps close	Correct signal $\rightarrow$ preferences adjust $\rightarrow$ investment rises endogenously
Primary target	Budget constraint (w_t , income, assets)	Signal factor (λ) and preference dynamics (θ)
Instrument	Transfers, quotas, earmarked programs	Information correction, market access, institutional reform
Success metric	Outcome gaps (income, enrollment, representation)	Signal gaps (perceived returns, aspiration levels, λ)
Time horizon	Project cycle (3–7 years)	Generational (~ 4 half-lives ≈ 15 –25 years at $\rho = 0.85$)
Welfare benchmark	Revealed preferences (adapted θ^L)	Counterfactual baseline ($\bar{\theta}$)
Spatial design	Spread across sectors and regions	Concentrate to cross tipping points

The table reveals a fundamental asymmetry. Standard mainstreaming measures its own success using a metric—revealed preferences—that the adaptive framework shows is itself distorted by the problem being addressed. A program that marginally raises women’s income without changing the signal structure will register as successful by standard evaluation while leaving the underlying preference gap untouched. Signal-based mainstreaming measures what actually drives long-run change: whether women’s perceived returns, aspiration levels, and investment willingness are converging toward the male benchmark.

A second asymmetry concerns the logic of concentration versus diffusion. Standard mainstreaming, operating on the budget constraint, follows a logic of equity: spread resources to as many women as possible in as many sectors as possible. Signal-based mainstreaming, operating on a dynamical system with tipping points, follows a logic of phase transition: concentrate effort where it can push the system into a self-sustaining virtuous cycle. These are incompatible design principles, and institutions that try to do both will default to the diffusion logic because it is easier to justify bureaucratically.

7 Sectoral Applications

To illustrate how signal-based mainstreaming operates in practice, I develop detailed applications in six sectors. For each, I trace the signal channel, contrast the standard and signal-based approaches, and identify the key evaluation metrics. The goal is not to provide program blueprints—those require context-specific design—but to show how the same analytical framework applies across domains.

Before turning to individual sectors, however, it is essential to address a cross-cutting signal channel that amplifies $\lambda < 1$ in every domain: the care economy.

7.1 The Care Economy as Cross-Cutting Signal Amplifier

Unpaid care work is the single largest category of gendered time allocation and arguably the most powerful signal channel in the entire economy. Women perform approximately 75% of the world’s unpaid care work, amounting to an estimated 12.5 billion hours per day globally (Ferrant et al., 2014; Charmes, 2019). Standard mainstreaming has acknowledged this gap but has struggled to address it, largely because the resource-based framework lacks the tools to analyze an allocation problem that operates primarily through signals and preferences rather than through market prices. Conventional responses—time-use surveys, public provision of childcare, campaigns to encourage men’s domestic participation—are resource-side interventions: they change the time and infrastructure available to women but do not address the signal structure that produces the gendered allocation in the first place.

The care economy matters for this paper’s argument because it operates as a *signal multiplier* across all other sectors. A girl who grows up observing that her mother spends five hours per day on unpaid domestic work while her father spends one hour receives a composite signal about the return to female investment: whatever the labor market pays women, the effective return to market work is reduced by the care burden that she expects to bear. This is not a separate signal from the labor market signal—it is the same λ , operating through a different channel. The care burden reduces the effective return to every form of female investment: education (fewer hours for study), career (interrupted trajectories), entrepreneurship (less time for business development), and financial saving (lower disposable income after care costs). In terms of the framework, the care burden enters as a multiplicative discount: the signal a woman receives is not λr_t but $\lambda_{\text{sector}} \times \lambda_{\text{care}} \times r_t$, where $\lambda_{\text{care}} < 1$ reflects the expected care burden. This means that every sectoral signal channel documented below is *amplified* by the care economy—a point that standard mainstreaming, which treats care as a separate policy domain, systematically misses.

Critically, the care allocation is itself endogenous to preferences that are themselves

endogenous to the care allocation. A woman with adapted preferences ($\theta^L < \bar{\theta}$) values future consumption less and therefore invests less in her career; lower career investment means lower market earnings; lower market earnings make her time relatively cheaper in home production; and the household optimally allocates more of her time to care work. The care burden is both a cause and a consequence of adapted preferences—a tight feedback loop within the broader $\lambda \rightarrow \theta \rightarrow s \rightarrow K \rightarrow r \rightarrow \lambda$ cycle.

Signal-based interventions for the care economy operate through three mechanisms. First, policies that raise the shadow price of women’s time—not through subsidies but through raising the market return to female labor (which depends on the sector-specific interventions below). When women’s market earnings rise, the household’s optimal time allocation shifts. Second, social norms interventions that target men’s care participation, not through “awareness campaigns” (which are resource-side) but through visibility interventions that change the signal about male care work: men in the community observing other men performing care work without status loss. This is the aspiration-shock mechanism applied to masculinity norms. Third, infrastructure investment (public childcare, elder care) framed not as “freeing women’s time” (a resource frame) but as eliminating a signal distortion: making the return to female market work visible without the care-burden discount.

With this cross-cutting amplifier in mind, I turn to six sector-specific applications.

7.2 Agriculture

Agriculture accounts for the largest share of female employment in low-income countries—over 60% in Sub-Saharan Africa and more than 40% in South Asia ([World Bank, 2012](#)). Gender gaps in agricultural productivity are well-documented: the FAO has estimated that women farmers achieve 20–30% lower yields than men on comparable plots, a gap largely attributable to differential access to inputs, extension, and markets rather than to differences in intrinsic ability ([FAO, 2011](#)).

Standard mainstreaming. The conventional response targets the resource gap directly: allocate land titles to women, provide female-targeted extension services, distribute improved seeds and fertilizer, establish women’s cooperatives for input procurement. Programs are evaluated on hectares cultivated by women, yields, and income. The theory of change is straightforward: give women the same inputs as men, and productivity will converge.

This approach has achieved real gains in input access but has struggled to close the productivity gap. A systematic review by [Kilic et al. \(2015\)](#) found that even after controlling for observable input differences, a significant gender productivity gap persists in many African

contexts. The standard interpretation is that unobservable constraints—time poverty, labor availability, social norms—explain the residual. The signal interpretation is different.

The signal channel. Women farmers face at least three sources of signal discount. First, gender-segmented output markets: in many contexts, women sell through different intermediaries than men, often at lower prices. Aggarwal (2018) documented significant gender price gaps in Ugandan agricultural markets, with women receiving 4–8% less for identical crops. The signal is not the crop’s potential yield but the *return on investment in improved practices*: if a woman farmer expects to receive a lower price for higher-quality output, her perceived return to quality investment is λr with $\lambda < 1$, and her optimal response is to invest less—which standard mainstreaming will record as “lower adoption of improved practices.” Second, extension systems operate primarily through male-dominated farmer networks. Kondylis et al. (2016) showed that conventional farmer-to-farmer extension in Mozambique largely bypassed women, not because women were excluded from training but because the contact farmers who disseminated information were overwhelmingly male, and diffusion followed gendered social networks. Women literally received a weaker informational signal about the returns to new technologies. Third, contract farming—the highest-return channel in many agricultural value chains—is overwhelmingly male. Maertens and Swinnen (2012) found that in Senegalese horticulture, women were largely excluded from contract arrangements with export firms, not by explicit policy but through the combination of land title requirements, minimum volume thresholds, and male-mediated access to buyer relationships.

Signal-based intervention. The reorientation targets λ at each point in the value chain. Digital market platforms that display real-time prices and enable direct buyer matching eliminate the intermediary who imposed the gender price discount (Aker, 2016). Demonstration plots managed *by women* in women’s networks (not gender-mixed groups where men dominate discussion) correct the extension signal by showing returns under female management conditions. Contract farming programs designed with lower minimum volumes and group-based contracts that allow women’s cooperatives to aggregate output address the structural exclusion from high-return channels.

The tipping-point logic applies directly: if women’s participation in high-return market channels is sufficiently visible, it changes the signal for the next cohort of girls making educational and occupational decisions. The key design choice is *saturation*—enough women in enough markets within a given community that the signal change is observable to adolescents.

Evaluation shift. Standard: yields, income, input use. Signal-based: gender gap in realized farm-gate prices; gender gap in awareness of market prices and new technologies (measured by survey); women’s perceived return to adopting improved practices; aspiration levels of girls in farming households regarding occupational expectations.

7.3 Financial Inclusion

The microfinance revolution of the 1990s and 2000s was explicitly framed as a gender mainstreaming strategy: by providing credit to women, it would unlock women’s entrepreneurial potential. Decades later, the evidence is more nuanced. Systematic reviews of microfinance impacts ([Banerjee et al., 2015b](#); [Meager, 2019](#)) find modest effects on business income and no consistent effects on women’s empowerment, consumption, or education. This is puzzling under a resource-constraint model—credit should enable investment, which should raise returns. It is less puzzling under a signal model.

Standard mainstreaming. Microfinance for women, women’s savings groups, gender-targeted credit lines, mobile money wallets, financial literacy training. Evaluate: loans disbursed, savings balances, repayment rates, number of women with accounts.

The signal channel. Financial products available to women frequently offer worse terms than those available to men, creating a literal return discount. [Demirgüç-Kunt et al. \(2013\)](#) documented that in many developing countries, women face higher collateral requirements, shorter loan maturities, and higher effective interest rates, even controlling for risk profiles. In microcredit, group lending imposes implicit costs (meeting time, joint liability risk) that reduce the net return relative to the individual loans typically available to male borrowers. More subtly, women are overwhelmingly concentrated in low-return financial products: basic savings accounts and microloans for petty trade, rather than term deposits, investment funds, or business loans for growth-oriented enterprises. A woman whose entire financial experience consists of a 2% savings account and a microloan at 30% annual interest is receiving a very different return signal than a man whose experience includes equity investment and commercial credit at 15%.

The aspiration channel compounds the problem. When a woman’s reference group consists of other women using basic financial products for consumption smoothing, her perception of what financial investment *can do* is shaped by that low-return experience. The signal is not “you could earn a high return if you invested more”; the signal is “financial products for people like you yield modest returns,” and the rational adaptive response is modest saving.

Signal-based intervention. Three distinct mechanisms. First, regulatory reform that eliminates gendered pricing: mandating equal terms for equal risk profiles, prohibiting collateral requirements that disproportionately exclude women (who typically lack titled land), and requiring financial institutions to report gender-disaggregated product data including effective interest rates and returns. This is a direct λ -raising intervention that operates through the institutional structure of financial markets. Second, product design that exposes women to higher-return instruments: investment clubs that pool capital to access equity markets, matched savings programs that demonstrate compounding returns, and “graduation” programs that move women from basic savings to diversified investment portfolios. The key is that the product itself generates a return signal: a woman who earns 8% on an investment club portfolio has received a fundamentally different signal than a woman earning 2% on a savings account, even if the capital invested is identical. Third, peer-group investment models that generate visible success stories within the community. If financial success by women is rare and invisible, it generates no signal change for the next cohort; if it is common and visible, it operates as an aspiration shock that can trigger the tipping-point dynamics.

Evaluation shift. Standard: number of accounts, savings volume, repayment rates. Signal-based: gender gap in effective interest rates and net returns across comparable products; women’s perceived expected return on financial investment (measured by survey instruments); share of women using higher-return products (investment accounts, equity participation, business credit); intergenerational transmission of financial behavior (do daughters of treated women use financial products differently?).

7.4 Education

The education sector illustrates the resource-signal distinction with particular clarity because the resource approach has achieved its most visible successes here—and precisely those successes reveal its limitations. Gender gaps in primary enrollment have narrowed dramatically across the developing world over the past three decades. In most countries, girls now attend primary school at rates comparable to boys, and in many contexts the gender gap has reversed at the secondary level ([World Bank, 2012](#)). Yet progress in enrollment has not translated proportionally into progress in the labor market: women’s labor force participation, wage returns to education, and representation in high-return occupations have improved far less than the enrollment statistics would predict ([Duflo, 2012](#)). Standard mainstreaming, having solved the input problem (access to school), now confronts the signal problem (what school is perceived to be worth).

Standard mainstreaming. Girls’ scholarships, conditional cash transfers tied to enrollment, school construction in underserved areas, elimination of school fees, provision of sanitary facilities to reduce dropout. Evaluate: enrollment, attendance, grade completion, test scores.

The signal channel. The return to education is a composite: years of schooling $\times$ quality of learning $\times$ return to human capital in the labor market. Standard mainstreaming targets the first component (enrollment) and sometimes the second (learning quality). But the signal that drives aspiration formation and investment decisions is primarily the third: what educated women *earn* relative to uneducated women, and relative to educated men. If the labor market discounts women’s qualifications—through occupational segregation, wage discrimination, or normative penalties for female employment in high-return sectors—the perceived return to schooling is λr where r is the male return.

Jensen (2012) demonstrated this mechanism directly in an Indian context. Providing young women with accurate information about the returns to education in the business process outsourcing (BPO) sector—returns that were substantially higher than what their communities perceived—significantly increased enrollment and delayed marriage. The intervention provided no resources: no scholarships, no infrastructure, no cash transfers. It corrected the signal. Conversely, Attanasio et al. (2017) found that a vocational training program in Colombia had limited long-run effects on women’s employment, partly because the occupations women were trained for offered returns that were lower than expected—the signal *after* training was discouraging, undoing the aspiration effect of enrollment.

A particularly important signal mechanism in education operates through the *curriculum* and *peer composition*. Girls in gender-segregated STEM tracks who see other girls succeed in technical subjects receive a different signal about the return to technical investment than girls in mixed settings where boys dominate technical performance and teacher attention. Booth and Nolen (2012) found that girls in all-female experimental groups in university economics were significantly more likely to choose competitive options, suggesting that the peer-composition signal affects risk preferences directly.

Signal-based intervention. Three tiers. Tier one (signal correction): information campaigns providing accurate, context-specific data on returns to female education in non-traditional occupations. This follows the Jensen model but must be sustained and updated, not a one-shot intervention, because the signal environment—the labor market—continues to generate discouraging signals that the information must counteract. Tier two (aspiration shock): structured mentoring programs that connect girls to women in high-return careers,

following the logic of [Beaman et al. \(2012\)](#). The design principle is *visibility*: the mentor’s success must be observable to the girl’s community, not just to the girl herself, so that the signal change affects the entire reference group. Tier three (structural λ -raising): labor market reforms that increase the actual return to female education—equal pay mandates, anti-discrimination enforcement, affirmative action in high-return sectors. Without this tier, the first two produce raised aspirations that the labor market then crushes, generating a particularly damaging signal: “even when you invest, it doesn’t pay off.” The sequencing matters: tier three must be in place (or credibly committed) before tiers one and two, or the intervention risks aspiration frustration—a phenomenon [Genicot and Ray \(2017\)](#) formalized as the aspirations gap becoming too large to be motivating.

Evaluation shift. Standard: enrollment, test scores, completion rates. Signal-based: girls’ perceived return to education (earnings expectations for women with secondary/tertiary degrees); occupational aspirations (what job do you expect to have at age 30?); field-of-study choices (share of girls entering high-return tracks); the aspiration transmission coefficient (do daughters of program participants have different educational expectations than daughters of non-participants?).

7.5 Labor Markets

Labor markets are where the signal structure is both most consequential and most visible. The return to labor market investment—education, experience, job search, skill acquisition, networking, career risk-taking—is directly observable through wages and career trajectories. Gender wage gaps, glass ceilings, and occupational segregation are therefore not just outcomes to be corrected; they are the *signal infrastructure* through which the next generation’s preferences are formed.

Standard mainstreaming. Quotas for female representation on boards and in management, anti-harassment policies, parental leave mandates, equal pay legislation (often weakly enforced), gender budgeting in public employment. Evaluate: share of women in management, gender pay gap, labor force participation rate.

The signal channel. At least four signal mechanisms operate simultaneously in labor markets. First, the *glass ceiling signal*: women observe that female career trajectories plateau earlier and at lower levels than male trajectories, even conditional on equivalent qualifications and experience. This truncation reduces the perceived lifetime return to career investment—not just the pecuniary return but the status and autonomy returns that drive ambition

(Goldin, 2014; Bertrand, 2018). Second, the *normative penalty signal*: women who display ambition, negotiate aggressively, or prioritize career over family face social costs that men do not. Bursztyn et al. (2017) showed that single women in a competitive MBA program strategically reduced their stated career ambitions when those statements were observable to male peers. The net return to ambition is $r - c$ for men and $\lambda r - c'$ for women, where $c' > c$ represents the higher social penalty. Third, the *network exclusion signal*: professional networks that control access to information about promotion opportunities, compensation benchmarks, and career strategies are disproportionately male. Women who lack access to these networks make career decisions based on less information, which is equivalent to receiving a noisier and on average lower signal about the returns to strategic career moves. Fourth, the *motherhood signal*: women observe that having children imposes a career penalty (the “child penalty”) that men do not face. Kleven et al. (2019) documented that the birth of a first child causes a 20% long-run earnings decline for women and near-zero decline for men across several developed countries. This gender-specific signal operates powerfully on the next generation: girls who observe their mothers’ career trajectories stalling after childbirth receive a discounted return signal for career investment relative to family investment.

Signal-based intervention. The critical insight is that conventional labor market policies—quotas, parental leave, anti-harassment measures—are primarily *access* interventions. They change who is in the room but not necessarily the return signals that people in the room observe. Pay transparency mandates are a genuine signal intervention: they make r directly observable to women, eliminating the information asymmetry component of the signal discount. Baker et al. (2023) found that pay transparency policies in Canada reduced gender wage gaps, suggesting that information correction has a direct effect on outcomes even without changes in formal rules.

Sponsorship programs—as distinct from mentoring—are a tipping-point intervention. Mentoring provides advice; sponsorship provides access, advocacy, and visible placement in high-authority roles. When a critical mass of women visibly occupy senior positions *and are seen to succeed*, the career return signal for the next cohort of women shifts. The design principle is visibility and credibility: a token female board member does not change the signal (and may worsen it if she is seen as a token); a critical mass of female leaders who visibly exercise authority and earn at parity does. The tipping-point logic implies a minimum threshold—likely around 30% representation at senior levels, consistent with the “critical mass” literature in political science (Dahlerup, 2006)—below which the signal change is insufficient to alter preference dynamics.

Policies that reduce the child penalty—publicly funded childcare, shared parental leave

with incentives for paternal uptake, workplace flexibility norms that apply to both genders—address the motherhood signal channel directly. These are λ -raising interventions that operate on one of the most powerful intergenerational signals: what girls observe happening to their mothers’ careers.

Evaluation shift. Standard: representation statistics, gender pay gap, labor force participation. Signal-based: women’s expected lifetime earnings profile (perceived career trajectory); perceived probability of reaching senior positions; willingness to negotiate and signal ambition (measurable through experimental choice tasks); perceived career cost of motherhood; daughters’ career aspirations conditional on mothers’ labor market experience.

7.6 Health and Reproductive Autonomy

Health is typically treated as a separate domain from economic mainstreaming, but the adaptive preferences framework reveals that health decisions—particularly those related to fertility, contraception, and bodily autonomy—are deeply embedded in the same signal structure.

Standard mainstreaming. Supply-side interventions: build clinics, train midwives, distribute contraceptives, provide prenatal care. Demand-side interventions: conditional cash transfers for health facility utilization, community health worker visits, information campaigns about family planning. Evaluate: maternal mortality, contraceptive prevalence, antenatal care visits, facility-based delivery.

The signal channel. Fertility decisions are investments: they involve a tradeoff between current costs (health risks, forgone earnings, care time) and perceived future returns (children’s economic contributions, old-age support, social status). The signal structure is gendered in a specific way: the *costs* of fertility fall disproportionately on women, but the *returns* are often controlled by men (through patrilineal inheritance, sons’ preference, male control of household resources). The signal a woman receives about the return to her reproductive investment is therefore discounted by the degree to which she controls the returns from that investment.

More subtly, women’s health-seeking behavior is itself shaped by adapted preferences. [Sen \(1990\)](#) observed that women in settings with strong son preference underreport their own illness and underinvest in their own health relative to male household members—not because health services are unavailable but because adapted preferences lead women to value their own health less. The signal channel operates through the intrahousehold allocation of

care: girls who grow up observing that mothers deprioritize their own health absorb a signal about the return to female health investment.

Contraceptive use illustrates the dynamic. Standard mainstreaming treats low contraceptive uptake as an access problem (supply-side) or an information problem (knowledge about methods). But in many contexts, women who know about and can access contraception choose not to use it—a choice that revealed-preference analysis interprets as a genuine preference for larger families. The signal interpretation is different: in settings where women’s status, security, and material wellbeing depend on having children (particularly sons), the perceived return to fertility-limiting investment is low or negative. The “preference” for large families is a rational adaptation to a signal structure in which women’s returns are contingent on children.

Signal-based intervention. Interventions that change the signal structure of fertility and health decisions. Programs that increase the return to female education and employment (Sections 6.3–6.4) change the opportunity cost signal: as the perceived return to non-reproductive investment rises, the relative attractiveness of early and frequent childbearing falls. [Jensen \(2012\)](#) found that information about labor market returns to education delayed marriage and reduced fertility—a signal intervention with demographic consequences. Cash transfer programs that *delink* women’s economic security from fertility—old-age pensions for women, property rights for widows, inheritance rights for daughters—change the signal about the return to investing in children versus investing in oneself. Community-level interventions that make low-fertility, high-investment families visible and successful change the reference-group signal, operating through the same aspiration mechanism as [Bernard et al. \(2014\)](#).

Evaluation shift. Standard: contraceptive prevalence, maternal mortality, antenatal visits. Signal-based: women’s perceived opportunity cost of childbearing; women’s control over returns from children (decision-making over children’s earnings, inheritance, old-age support); women’s self-valuation of own health (measurable through stated willingness-to-pay for health inputs or time-allocation diaries); girls’ observed intrahousehold health investment patterns.

7.7 Digital Economy and Entrepreneurship

The digital economy represents a particularly important frontier for signal-based mainstreaming because it is simultaneously a domain of rapid growth, large gender gaps, and

powerful signal infrastructure—the algorithms and platforms that increasingly mediate economic information.

Standard mainstreaming. Digital literacy programs for women, women’s coding bootcamps, female entrepreneurship funds, incubators with gender quotas. Evaluate: number of women trained, startups founded by women, women’s share of digital sector employment.

The signal channel. Digital markets generate signals with unprecedented reach and specificity, and those signals are systematically gendered. [Lambrecht and Tucker \(2019\)](#) showed that algorithms trained on behavioral data will optimize toward displaying high-paying career ads to men and lower-paying ads to women—not because the algorithm was programmed to discriminate but because gendered behavioral patterns in the training data lead the optimizer to a discriminatory equilibrium. This is $\lambda < 1$ operating at algorithmic scale: millions of women receive discounted return signals about career opportunities every time they browse the internet.

In entrepreneurship, the signal channel operates through funding. Female founders receive a smaller share of venture capital (approximately 2% of total VC funding in the United States), face different questions in pitch meetings (“prevention”-oriented questions about risk, rather than “promotion”-oriented questions about growth, as documented by [Kanze et al. 2018](#)), and are offered worse terms conditional on receiving funding. A woman contemplating a startup observes this signal environment and rationally adjusts her aspiration: the perceived return to entrepreneurial investment is λr with λ substantially below 1. Standard mainstreaming responds by creating women’s funds and incubators, which address the budget constraint but do not change the broader signal environment—indeed, they may reinforce the perception that women need special help, which is itself a discouraging signal.

Platform economies generate a third signal mechanism: ratings and reviews. [Edelman et al. \(2017\)](#) documented discrimination against minority hosts on Airbnb; similar patterns have been found for female service providers on freelancing platforms. If women systematically receive lower ratings for equivalent work, the signal about the return to quality investment on the platform is discounted, and women rationally invest less in platform-based careers.

Signal-based intervention. Algorithmic auditing mandates that require platforms to test for and correct gendered signal distortion in advertising, search results, and opportunity display. This is the 21st-century analog of pay transparency: making the return signal visible and equal. For entrepreneurship, the signal intervention is structural reform in funding: requiring investors to report gender-disaggregated funding data (signal transparency), training

investors to ask equivalent questions of male and female founders (signal correction), and creating co-investment vehicles that *de-risk* female-led ventures for mainstream investors rather than segregating them into women-only funds (integration rather than earmarking). For platform economies: blind review systems, audit requirements on rating disparities, and platform design that makes female success visible to other women on the platform (aspiration feedback).

The tipping-point logic is especially powerful in digital markets because network effects amplify signals. If a critical mass of visible female entrepreneurs succeeds on a platform, the algorithmic signal itself shifts: recommendation engines trained on diverse success stories will generate more equitable signal distributions for the next cohort.

Evaluation shift. Standard: number of women trained, startups launched, digital sector employment share. Signal-based: gender gap in algorithmic ad exposure for high-return opportunities; gender gap in VC pitch offer terms conditional on venture quality; platform rating gaps conditional on quality metrics; women’s perceived expected return to entrepreneurship and digital career investment.

8 Conclusion

Gender mainstreaming has operated for three decades on the budget constraint: disaggregate, transfer, add a gender component. This paper has argued, drawing on practitioner critique, feminist economics, behavioral evidence, and formal adaptive preference theory, that the binding constraint is informational. Women receive systematically discounted return signals that reshape their preferences across generations, producing inequality that appears voluntary but is endogenous to the signal structure. Transfers cannot fix signal problems. Incremental reforms cannot cross tipping points. Short programs cannot outlast hysteresis. And welfare evaluation using adapted preferences cannot detect the true cost of inequality.

The sectoral analysis—spanning agriculture, finance, education, labor markets, health, the digital economy, and the care economy—reveals a common pattern. In every domain, standard mainstreaming targets the resource gap while leaving the signal structure intact. In agriculture, women receive inputs but still face gender-segmented output markets that discount their returns. In finance, women receive microloans but remain concentrated in low-return products that generate discouraging signals. In education, girls receive scholarships but observe a labor market that pays educated women less than educated men. In labor markets, quotas place women in positions but do not change the career trajectory signals

that shape the next generation’s aspirations. In health, contraceptives are supplied but the signal structure that links women’s security to fertility remains unaddressed. In the digital economy, women are trained but algorithmic systems reproduce and amplify the signal discount at scale. And across all sectors, the care economy operates as a signal amplifier that reduces the effective return to every form of female investment.

The alternative—signal-based mainstreaming—requires a reorientation of the entire enterprise. Not from “more resources for women” to “fewer resources for women,” but from the question “how do we give women more?” to the question “how do we change what women see when they look at the returns to investing in themselves?” This reorientation is not a discovery of new facts—the practitioner critique, the philosophical foundations, and the behavioral evidence have each been developing for decades. What it offers is a formal mechanism that unifies those strands into a single analytical framework, proves that the standard policy toolkit is structurally inadequate (not merely poorly implemented), and derives operational principles that follow from the structure of the problem rather than from intuition alone. The five principles—map the signal channel, target λ , design for tipping points, evaluate against counterfactual preferences, and embed the feedback loop—provide a framework for this reorientation. They demand new diagnostic tools (signal audits rather than gender checklists), new metrics (perceived returns and aspiration levels rather than enrollment counts and income measures), new time horizons (generational rather than project-cycle), and a new design logic (concentration for phase transition rather than diffusion for coverage).

The stakes are not merely academic. If the adaptive preferences framework is correct, the current approach to gender equality is not merely slow—it is structurally incapable of reaching its objective. It targets the wrong channel. And every generation spent targeting the wrong channel is a generation in which preferences adapt further, the trap deepens, and the tipping point recedes. The cost of policy misdiagnosis is not stasis; it is entrenchment.

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